9.2: Norms

The norm of a vector in an arbitrary inner product space is the analog of the length or magnitude of a vector in $\mathbb{R}^n$. We formally define this concept as follows.

Definition 9.2.1. Let V be a vector space over $\mathbb{F}$. A map

$$\begin{aligned} \|\cdot\| : V &\to \mathbb{R} \\ v &\mapsto \|v\| \end{aligned}$$

is a **norm** on V if the following three conditions are satisfied.

1. **Positive definiteness**: $\|v\| = 0$ if and only if $v = 0$;
2. **Positive Homogeneity**: $\|av\| = |a|\,\|v\|$ for all $a \in \mathbb{F}$ and $v \in V$;
3. **Triangle inequality**: $\|v + w\| \le \|v\| + \|w\|$ for all $v, w \in V$.

Remark 9.2.2. Note that, in fact, $\|v\| \ge 0$ for each $v \in V$ since

$$0 = \|v - v\| \le \|v\| + \|-v\| = 2\|v\|. \tag{9.2.1}$$

Next we want to show that a norm can always be defined from an inner product $\langle \cdot, \cdot \rangle$ via the formula

$$\|v\| = \sqrt{\langle v, v\rangle} \text{ for all } v \in V. \tag{9.2.1}$$

Properties 1 and 2 follow easily from Conditions~1 and 3 of Definition~9.1.1. The triangle inequality requires more careful proof, though, which we give in Theorem~9.3.4??? in the next chapter.

If we take $V = \mathbb{R}^n$, then the norm defined by the usual dot product is related to the usual notion of length of a vector. Namely, for $v = (x_1, \ldots, x_n) \in \mathbb{R}^n$, we have

$$\|v\| = \sqrt{x_1^2 + \cdots + x_n^2}. \tag{9.2.2}$$

We illustrate this for the case of $\mathbb{R}^3$ in Figure 9.2.1.

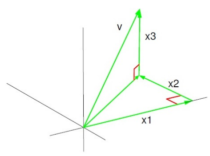

Figure 9.2.1: *The length of a vector in* $\mathbb{R}^3$ *via equation* 9.2.1.

While it is always possible to start with an inner product and use it to define a norm, the converse requires more care. In particular, one can prove that a norm can be used to define an inner product via Equation 9.2.1 if and only if the norm satisfies the Parallelogram Law (Theorem 9.3.6~???).

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